AMA572 Lecture 2.

Credit Risk (II): Contingent Claims

1 Contingent Claims

Contingent claims are claims that can be made only if one or more specified outcomes occur. In other words, the payoffs of a contingent claim depend on the values of other assets or numerical indices. Obvious examples of contingent claims are: put and call options, warrants, futures contracts, and convertible bonds. In addition, as we will see in this section that the value of a firm's debt and equity depends on the value of the firm's assets, and even taxes paid to the government can be viewed as contingent claims.

Suppose the firm's asset value is $V(t)$. Assume that the firm has only two types of claims: equity and debt. More precisely, we assume that the firm issue a bond with face value D with maturity T. Assume (i) there is no coupon payment; (ii) there is no tax credit to coupon payments; (iii) there is no bankruptcy cost. Also, for simplicity, we assume that at time T the firm will liquidate the asset and deliver all the remaining asset, after paying the debt holders, to the equity holders.

Then the total payment to the debt holders at time T is

$$\min(V(T), D).$$

Similarly the total payment to the equity holders at time T is

$$(V(T) - D)^+.$$

Thus, both debt and equity can be viewed as contingent claims. In particular, the equity payoff is the same as that a call option with strike price D and the underlying asset V.

The above analysis can also be extended to subordinated bonds. More precisely, suppose there are two classes of debt, senior debt and junior debt. In the event of default the senior debt holders will collect the money first, and then the junior debt holders, and whatever left if any will finally go to the equity holders. Suppose the face values of the senior and junior debt are D_S and D_J respectively. Then we have the following payoff function.

	$V_T < D_S$	$D_S \le V_T < D_S + D_J$	$D_S + D_J \le V_T$
Senior	V_T	D_S	D_S
Junior	0	$V_T - D_S$	D_J
Equity	0	0	$V_T - (D_S + D_J)$

Therefore, the equity has a payoff $\{V_T - (D_S + D_J)\}^+$ that is identical to a call option with strike price $(D_S + D_J)$. The senior debt has a payoff

$$V_T - (V_T - D_s)^+,$$

and the junior debt has a payoff

$$(V_T - D_s)^+ - \{V_T - (D_S + D_J)\}^+ .$$

2 Optimal Capital Structure and Modigliani-Miller Theorem

The contingent claim approach to value debt and equity values is also useful to study corporate finance. One of the most famous results in corporate finance is Modigliani-Miller Theorem (MM theorem), which attempts to understand the optimal capital structure of a firm. The capital structure problem means that, when a firm needs capital, it has to decide how much of the capital should be financed by issuing debts and how much by issuing equity.

Here is a simplest possible version of the MM theorem. Assume the market is perfect and is in equilibrium. In this simple setting the decision of the capital structure becomes the choice of the face value of the debt D.

The total market value of equity is

$$S(0) = E^*[e^{-rT}(V(T) - D)^+],$$

where the expectation E^* is taken under some risk-adjusted probability P^*. Similarly the total market value of the debt will be

$$B(0) = E^*[e^{-rT} \min(V(T), D)].$$

Thus, the total market value of firm, $M(0)$, which is defined to be the sum of the market values of debt and equity, is given by

$$\begin{aligned} M(0) &= S(0) + B(0) \\ &= E^*[e^{-rT}(V(T) - D)^+] + E^*[e^{-rT} \min(V(T), D)] \\ &= E^*[e^{-rT} \left\{(V(T) - D)^+ + \min(V(T), D)\right\}] \\ &= E^*[e^{-rT} V(T)], \end{aligned}$$

which is independent of D.

In other words, since the total market value of the firm is independent of D, the firm's capital structure is irrelevant for the market value of the firm!

This is a simple yet very surprising conclusion, which leaded to a Nobel Prize for MM.

Note that in above argument, we never use the "risk-neutral" assumption; more precisely, we never assume that $E^*[e^{-rT}V(T)] = V(0)$. As long as we can pricing contingent claims by using a "risk-adjusted" probability, the argument works. For further details of how to derive the "risk-adjusted" probability from equilibrium argument, see Kou (2002, Management Science), in which the risk-adjusted probability is derived explicitly for a general model (including model with jumps).

Also, it worth to pointing out that in the original MM paper published in 1958, they used an arbitrage argument rather than the equilibrium argument as above. More precisely, they assumed that there are two identical firms (of course this is a big assumption), with one firm having only equity, and the other one having both debts and equity. Then it is very easy to see that arbitrage argument between the two firms will prove that the MM result. In fact, we can treat the all-equity firm as the underlying asset for the other firm and use the option pricing arbitrage argument.

Of course, the empirical evidences suggest the totally opposite conclusion: the capital structure is relevant for firm's finance decisions. More precisely, here are some main empirical findings.

- Higher volatility leads to lower debt/equity ratio.
- Higher recovery value leads to higher debt/equity ratio.
- More maturity companies tend to have higher debt/equity ratio.
- High tech companies tend to have no debts.

To explain these empirical facts, one can look at the trade-off between the bankruptcy costs and tax credit. Let us begin with a simple example. Suppose the bond pays no coupon, and the principle value of the debt is D to be paid at maturity T, if there is no bankruptcy at time T, and the tax rate is κ. Thus, κD is the tax credit if there is no bankruptcy at time T. Note that for U.S. companies the debt payments (both the interests and principle payments) are tax deductible.

Assume that upon bankruptcy the firm losses $(1 - \alpha)$ of its asset value. This is because the bankruptcy will incur legal costs (such as lawyer fees), financial costs (difficulty in borrowing money), operational costs (e.g. damage to the brand name, layoff employees). Note that the bankruptcy occurs whenever $V(T)$ is less than D. Now the debt and equity values are given by

$$B(0) = E^*[e^{-rT}(D \cdot I\{V(T) \geq D\} + e^{-rT}\alpha V(T) \cdot I\{V(T) < D\}],$$

$$S(0) = E^*[e^{-rT}(V(T) - D + \kappa D) \cdot I\{V(T) \geq D\}].$$

The tax credit is $A(0)$ is given by

$$A(0) = E^*[e^{-rT}\kappa D \cdot I\{V(T) \geq D\}]. \tag{1}$$

The bankruptcy cost $C(0)$ is given by

$$C(0) = E^*[e^{-rT}(1-\alpha)V(T) \cdot I\{V(T) < D\}]. \tag{2}$$

Then it is easy to see that

$$\begin{aligned} M(0) &= B(0) + S(0) \\ &= E^*[e^{-rT}(V(T) + \kappa D) \cdot I\{V(T) \geq D\}] \\ &\quad + E^*[e^{-rT}\alpha V(T) \cdot I\{V(T) < D\}] \\ &= E^*[e^{-rT}V(T)] + E^*[e^{-rT}\kappa D \cdot I\{V(T) \geq D\}] \\ &\quad + E^*[e^{-rT}(\alpha - 1) \cdot V(T) \cdot I\{V(T) < D\}] \\ &= E^*[e^{-rT}V(T)] + A(0) - C(0). \end{aligned}$$

More precisely, the total market value of the firm is the sum of the asset value of the firm and the tax credit minus the bankruptcy cost:

$$M(0) = E^*[e^{-rT}V(T)] + A(0) - C(0).$$

Although the above derivation is based on a simple example, the conclusion that total market value of the firm is the asset value plus the tax credit minus the bankruptcy cost holds for more general case, including coupon payments, bonds with various maturities etc.

Note that $A(0)$ is an increasing function of D when D is small, and a decreasing function of D when D is large (as the bankruptcy chances becomes higher when D increases). $C(0)$ is an increasing function of D. In fact, it can be shown that $A(0) - C(0)$ increases in D first, and then decreases.

Now, if we want to choose D to optimize the total market value of the firm, i.e., maximizing $B(0) + S(0)$, then we realize that the optimal choice of D (or the optimal capital structure) is a balance of the trade-off between the tax credit of issuing bonds and the bankruptcy cost of issuing bonds.

More advanced models such as the Leland-Toft model or jump model in Chen and Kou (2006) can give a better explanation of the empirically observed capital structure.

Exercise 1. Compute (1) and (2) under the Black-Scholes-Merton geometric Brownian motion model.

3 Merton's Credit Model

3.1 Bond Price

In Merton's model an equity value is a call option on the corresponding asset value. There is no bankruptcy costs (i.e., $\alpha = 1$) and there is no tax credit (i.e., $\kappa = 0$).

We have the Black-Scholes formula

$$S(t) = V_t \Phi\left(d_1\right) - De^{-r(T-t)} \Phi\left(d_2\right), \tag{3}$$

where

$$d_1 = \frac{\log(V_t/D) + \left(r + \sigma_V^2/2\right)(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma_V\sqrt{T-t} \tag{4}$$

with σ_V is the volatility of the asset value. Therefore, in Merton's model, the bond value at time t is

$$\begin{aligned} B(t,T) &= V_t - C^{BS}(V_t, D, T-t, \sigma_V, r) \\ &= V_t - V_t\Phi(d_1) + D\exp(-r(T-t))\Phi(d_2) \\ &= V_t\Phi(-d_1) + D\exp(-r(T-t))\Phi(d_2), \end{aligned}$$

where $C^{BS}(V_t, D, T-t, \sigma_V, r)$ denotes the Black-Scholes formula for the call option.

Since under P^*, we have

$$V(T) = V(t)\exp\left\{(r - \frac{1}{2}\sigma^2)(T-t) + \sigma(W_T - W_t)\right\}.$$

Also, note that at time t the risk-adjusted default probability is given by

$$\begin{aligned} & P^*(V(T) < D) \\ = \; & P^*\left\{\frac{\log(V(T)/V(t)) - (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} < \frac{\log\left(D/V(t)\right) - (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}\right\} \\ = \; & P^*\left\{\frac{\log(V(T)/V(t)) - (r - \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}} < -d_2\right\} \\ = \; & \Phi(-d_2). \end{aligned}$$

3.2 Credit Spread

Hence the yield of the bond at time t is

$$\begin{aligned} y(t,T) &= \frac{1}{T-t}\log\frac{D}{B(t,T)} \\ &= \frac{1}{T-t}\log\frac{D}{V_t\Phi(-d_1)+D\exp(-r(T-t))\Phi(d_2)} \end{aligned}$$

and the credit spread in Merton's model is given by

$$\begin{aligned} s(t,T) &= y(t,T)-r \\ &= \frac{1}{T-t}\log\frac{D}{V_t\Phi(-d_1)+D\exp(-r(T-t))\Phi(d_2)}-r, \end{aligned}$$

where d_1 and d_2 are defined in equation (4).

In terms of the shape of the credit spread $s(t,T)$, meaning the curve of the credit spread the model cannot generate flexible credit spread similar to market observations. More precisely, there are only two shapes available for the credit spread under Merton's model.

(i) When $D < V(0)$ Merton's model generates humped curve with the initial credit spread at zero being 0.

(ii) When $D > V(0)$, the model leads to monotonically decreasing credit spreads with the credit spread at zero being ∞. ($D > V(0)$ is possible especially if the bond investor believe that the asset price will go up significantly so that the firm will have enough money to cover the debt obligation at the bond maturity). Of course, by making the maximum point of the humped curve sufficiently large, we can see a monotonically increasing credit spreads within a range of maturities typically observed. However, in practice, we see all three kinds of credit spreads, humped, increasing, decreasing for both investment grade bonds and speculative grade bonds.

3.3 Empirical Fitting

Empirically Merton's model does not fit the data well. From modelling perspective, there are several issues with the model.

(i) In terms of the shape of the credit spreads, the model cannot generate flexible credit spreads similar to market observations.

(ii) The model does not have coupon payments.

(iii) The model assumes that the default can only happen at the maturity of the bond.

However, Merton's model seems to predict the relative ranks of (both historical and implied) default probabilities reasonably well. That is why Moody uses Merton's model and monotone transforms to get historical default probabilities, and CreditGrades use the model and monotone transforms to get implied default probabilities.

The main advantage of Merton's model is its simplicity. More precisely, as long as one can price European call/put options, one can use the model. That is why it is widely used in practice. Other models, such as the first-passage-time models, are more complicated than Merton's model. But they can incorporate more empirical features.

4 Short Term Credit Spreads

One important empirical observation is that even for very short maturity bonds the credit spreads is still significant. This non-zero credit spreads for short maturity bonds cannot be produced easily by using models only based on diffusion stochastic processes with continuous sample paths, such as Brownian motion.

To understand this difficulty, note that for a diffusion process X_t with continuous sample path we must have

$$\frac{P(|X_{t+h} - X_t| \geq \varepsilon)}{h} \to 0. \tag{5}$$

Note that the bond price is given by

$$\begin{aligned} B(0,T) &= E^* \left[De^{-rT} I\{V(T) \geq D\} \right] + E^* \left[\alpha V(T) e^{-rT} I\{V(T) < D\} \right] \\ &= De^{-rT} P^*\{V(T) \geq D\} + e^{-rT} E^* \left[\alpha V(T) I\{V(T) < D\} \right], \end{aligned}$$

where the recovery rate $0 \leq \alpha < 1$. Therefore,

$$De^{-rT} \geq B(0,T) \geq De^{-rT} P^*\{V(T) \geq D\}.$$

Thus, we have the following inequalities for the yield

$$y(0,T) = \frac{1}{T} \log \frac{D}{B(0,T)} \geq \frac{1}{T} \log \frac{D}{De^{-rT}} = r,$$

and

$$\begin{aligned} y(0,T) &= \frac{1}{T} \log \frac{D}{B(0,T)} \\ &\leq \frac{1}{T} \log \frac{D}{De^{-rT} P^*\{V(T) \geq D\}} \\ &= -\frac{1}{T} \log \left(P^*\{V(T) \geq D\} \right) + r, \end{aligned}$$

the credit spread $s(0,T)$ is

$$0 \leq s(0,T) = y(0,T) - r \leq -\frac{1}{T}\log\left(P^*\{V(T) \geq D\}\right).$$

Thus, letting $T \to 0$ leads to

$$\begin{aligned}
\lim_{T\to 0} s(0,T) &\leq \lim_{T\to 0} -\frac{1}{T}\log\left(P^*\{V(T) \geq D\}\right) \\
&= \lim_{T\to 0} -\frac{1}{T}\log\left(1 - P^*\{V(T) < D\}\right) \\
&\leq \lim_{T\to 0} -\frac{1}{T}\log\left(1 - P^*\{|V(T) - V(0)| > V(0) - D\}\right) = 0,
\end{aligned}$$

if $V(0) > D$, because equation (5) implies

$$\lim_{T\to 0} \frac{1}{T} P^*\{|V(T) - V(0)| > V(0) - D\} = 0,$$

and

$$\lim_{x\to 0} \frac{\log(1-x)}{-x} = 1.$$

In summary, the short credit spread may approach to zero.

5 Using Merton's Model to Estimate Implied Default Probabilities from Equity Values

In the above equation (3), we have two unknowns, $V(0)$ and σ_V. To calibrate the model to market data, we need another equation in addition to (3). By Ito formula, since $S(t) = C(V(t), D, \sigma_V, r, T-t)$, we have

$$dS(t) = (\cdots)dt + \frac{\partial S}{\partial V} \cdot \sigma_V \cdot V(t) \cdot dW(t).$$

In particular, at time 0, the volatility σ_S of the return of $S(t)$ and the volatility of the asset return σ_V must satisfy

$$\sigma_S = \frac{\partial S}{\partial V} \cdot \sigma_V \frac{V(0)}{S(0)}.$$

Since $\frac{\partial S}{\partial V}$ is nothing but the delta in the Black-Scholes formula, we also have

$$\frac{\partial S}{\partial V} = \Phi\left(d_1\right). \tag{6}$$

In summary, we have two equations for two unknowns $V(0)$ and σ_V,

$$\begin{aligned}
S(0) &= V(0)\Phi\left(d_1\right) - De^{-rT}\Phi\left(d_2\right), \\
\sigma_S &= \Phi\left(d_1\right)\sigma_V \frac{V(0)}{S(0)}.
\end{aligned}$$

After solving $V(0)$ and σ_V, the implied (risk-neutral) default probability is simply

$$P^*(V(T) \leq D) = \Phi(-d_2). \tag{7}$$

Example. Suppose the equity value is \$3 million, and the equity return volatility $\sigma_S = 80\%$. The only debt has a maturity 1 year with par value \$10 million with no coupon payment. Assume $r = 5\%$. Then

$$S(0) = 3,\ \sigma_s = 0.8,\ r = 0.05,\ T = 1,\ D = 10.$$

Solving the two equations yields

$$V(0) = 12.4,\ \sigma_V = 0.2123.$$

Thus,

$$d_2 = 1.1408,$$

and the implied risk-neutral default probability is

$$\Phi(-1.1408) = 0.127.$$